

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

December 2013

CL103.01 Mechanics of Solids

Date: 07.12.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Locate the centroid of a lamina shown in Figure 1. [07]
- Q - 2 (a) Attempt any Two. [04]
- (i) Convert 2×10^5 MPa to kN/m^2 .
 - (ii) Two tensile forces 50kN and 40kN acting at a point with angle 60° between them. Find magnitude and direction of the resultant.
 - (iii) State principle of superposition.
- (b) Attempt any Two. [10]
- (i) Find the magnitude of two forces such that if they act at right angles, their resultant is $\sqrt{40}$ kN, but if they act at 60° , their resultant is $\sqrt{52}$ kN.
Note: sqrt: Square Root
 - (ii) Find the angle between two equal forces of magnitude P when their resultant is 1.5P. What is the maximum possible value of resultant?
 - (iii) Determine the resultant of a co-planner con-current force system if following forces are acting at a point:
 1. 150 kN tensile making an angle of 30° with +X axis.
 2. 200 kN tensile making an angle of 120° with +X axis.
 3. 80 kN tensile making an angle of 240° with +X axis.
 4. 200 kN tensile making an angle of 310° with +X axis.
- Q - 3 (a) A uniform ladder rests with one end on a horizontal floor and the other end against a vertical wall. Co-efficient of friction for the floor and the wall are 0.4 and 0.25 respectively. Find the inclination of the ladder when it is about to slip. [07]
- (b) Find forces in all members of a truss shown in Figure 2 or Figure 3. [07]

SECTION – II

- Q - 4 Derive $\Delta l = 4PL/\pi E d_1 d_2$ with usual notations. [07]
- Q - 5 (a) Explain stress-strain curve of a ductile material when subjected to a tensile loading up to failure. [04]
- (b) Attempt any Two. [10]
- (i) A circular bar 3m long, uniformly tapers from 50mm diameter to 25mm diameter. Find the magnitude of the force which will deform the bar by 0.8mm. $E = 200$ GPa.
 - (ii) A load of 2000kN is applied on a short concrete column 500mm x 500mm, reinforced with 4, 10mm diameter steel bars. Find stresses in steel and concrete if $E_s = 15 E_c$.

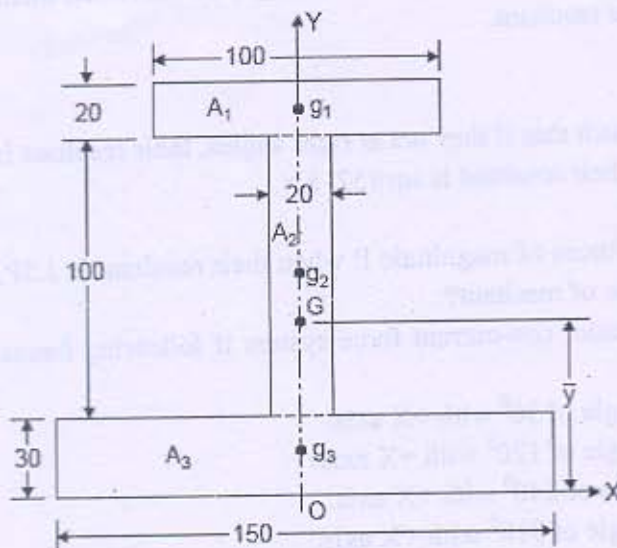
- (iii) A steel rod, 20mm diameter and 200mm long is heated through 100°C and at the same time subjected to an axial pull of 'P'. If the total extension of the rod is 0.30 mm, calculate the magnitude of pull 'P'. Consider $E = 200\text{GPa}$ and co-efficient of linear expansion is $12 \times 10^{-6}/^{\circ}\text{C}$.

Q - 6

Attempt any Two.

[14]

- (i) A man and a boy carry 30kg weight by means of a beam 2m long. Self weight of the beam is 10kg. Where the weight should be placed so that the man carries twice the weight carried by the boy?
- (ii) A beam ABC is 6m long. End A is hinged and end B is roller. Length of AB is 4m and BC is 2m. BC is an overhanging portion. Loads of 5kN and 8kN are acting at 1m and 3m from end A. Span BC is loaded with a UDL of 2 kN/m. Calculate the reactions at A and B.
- (iii) A cantilever beam of length 4m is subjected to a uniformly distributed load of 10kN/m over the entire span along with a point load of 10kN at the free end. Draws shear force and bending moment diagrams indicating values at all important points.



All dimensions in mm

Figure 1

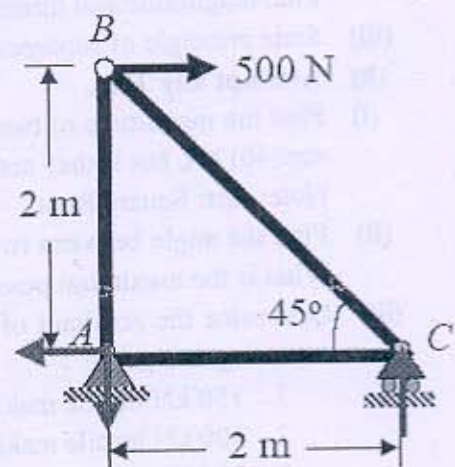


Figure 2

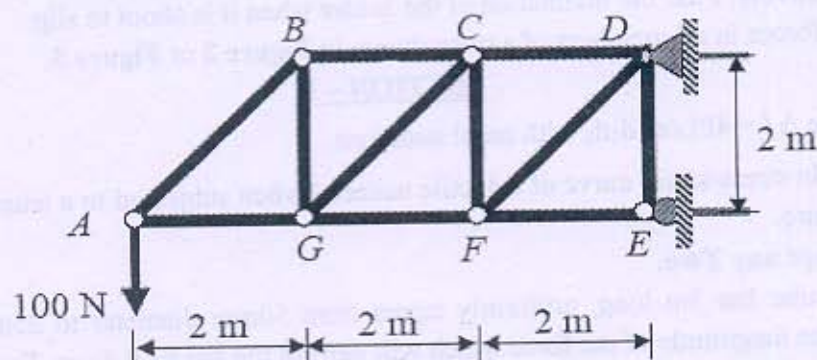


Figure 3
